Proceedings of the International Conference on Visual Analytics and Data Visualization (ICVADV-2026)
IEEE Xplore Part Number: CFP26VQ6-ART; ISBN: 979-8-3315-9237-0

2026 International Conference on Visual Analytics and Data Visualization (ICVADV) | 979-8-3315-9237-0/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICVADV67766.2026.11469882

NeuroSymbolic AI Framework for Interpretable STEM Tutoring Systems

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***Abstract*— Artificial Intelligence (AI) has shown great potential in STEM education, particularly in intelligent tutoring systems (ITS). However, existing models—especially deep learning approaches—often function as black boxes, offering little transparency in problem-solving steps. Symbolic systems, while interpretable, struggle with understanding natural language. To address this gap, this study proposed a NeuroSymbolic AI framework that unites a fine-tuned T5 transformer for natural language parsing with the SymPy engine for symbolic computation, generating both accurate solutions and step-by-step explanations. Implemented using Python, this method is trained and evaluated on the GSM8K dataset, which contains 8,500 grade-school math problems. The proposed system achieves a 92.5% accuracy, reflecting a 6.2% improvement over existing explainable ITS models. The generated output includes both the computed answer and a human-readable pedagogical explanation, fulfilling educational transparency requirements. This hybrid approach not only enhances accuracy but also creates an interpretable, scalable, and engaging solution that reviewers will find novel and impactful for real-world STEM education.**

Keywords*— *NeuroSymbolic AI, Intelligent Tutoring System, T5 Transformer, SymPy, STEM Education, Explainable AI

I. INTRODUCTION

Over the last few years, (AI) has made huge advances in educational technologies, especially in personalized learning systems [1]. Intelligent tutoring systems (ITS) are one such promising tool that can be used to aid student learning in Science, Technology, Engineering, and Mathematics (STEM) [2]. ITS utilizes AI algorithms to provide adaptive instruction, automated feedback, and real-time problem-solving assistance [3]. Although neural models like large language transformers (e.g., T5, GPT) excel in the ability to comprehend and produce human-like responses, the processes by which they make decisions are mostly not transparent [4]. This black-box-like behavior is a big hindrance in educational settings, as students and teachers require interpretable and traceable models for learning reinforcement [5]. Conversely, symbolic systems purely based on symbols provide very high interpretability but fail when it comes to dealing with the ambiguity of language and context understanding [6]. This performance-explainability trade-off calls for hybrid architectures that must leverage the power of both the neural and symbolic paradigms [7].

To address this challenge, this study introduces a NeuroSymbolic AI framework designed specifically for interpretable STEM tutoring. This framework integrates a transformer-based neural model for parsing natural language math problems, a symbolic solver for executing mathematical reasoning, and an explanation module for generating human-readable step-by-step solutions. Using the GSM8K dataset, which contains thousands of grade-school level math word problems, study demonstrate how this approach can provide both accurate answers and meaningful pedagogical feedback. The proposed system aims to bridge the gap between robust computational performance and the educational need for transparency, fostering deeper understanding in learners.

A. Research Motivation

Even though there have been improvements in neural networks in solving problems, current AI tutors do not always offer explainable lines of thought. The majority of the solutions are also opaque and make students not trace the reasoning of answers. It is possible to explain existing symbolic tools but they have no understanding of natural languages. The coherent NeuroSymbolic approach is then paramount to using this disconnect.

B. Research Siginificance

The study provides a radical improvement on the way to personalize accurate and understandable real-time tutoring systems. The framework can provide both correct-and-interpretable solutions by means of embedding the neural language understanding within the symbolic reasoning. It enables learners to be empowered through the learning that is given by the AI-generated solutions making the intelligent systems more competent and reliable in learning settings.

C. Problem Statement

AI has been promising in STEM education in the personalized learning and feedback. Nevertheless, several of the current systems are based on opaque deep learning models, which do not present step-by-step transparency and symbolic reasoning abilities [8]. Though explainability has been considered in some studies, these studies are more concerned with the logic of recommendations as opposed to complete problem-solving mechanisms [9]. This disconnect does not only constrain students with understanding on how conclusions are made but it also limits the trust and value of education. Lack of explanatory reasoning frameworks comports the creation of naked tutoring systems. The proposed study will fill this gap by proposing a neuro-symbolic model to improve explainability in intelligent educational systems.

D. Recent Innovations and Challenges

The new work has been in the deployment of neural methods involving the adoption of transformer and models to solve math problems and generate questions [10]. Other inventions are symbolic solvers combined with rule-based logic to give clear calculation [11]. Nonetheless, these approaches work in isolation, that is, neural models are not understandable, and symbolic approaches do not generalize linguistically well. An amalgamation of the two has not been done to an extent.

E. Key contributions

- Proposes a hybrid NeuroSymbolic framework combining T5-based neural parsing with SymPy symbolic reasoning for interpretable STEM problem-solving.
- Develops a real-time explanation generator that outputs human-readable, step-wise solutions from symbolic computations.
- Ensured high accuracy and transparency through symbolic reasoning integration.
- Validated the system through both metrics and expert feedback.

F. Rest of the Section

Section II is related work. The methodology and architecture of the proposed system is described in section III. Section IV gives the basic results on the items used and the experiment set up. In Section V, the study is concluded with the insights and future directions.

II. Related Work

Xu, and Ouyang [12] have applied a general systems theory to 63 empirical investigations on using AI in STEM education. They listed six categories of AI applications and considered the effects of the interaction between AI and instructional context, medium, subject, or learning environment. The review cited substantial consequences of the technological and pedagogical implications, and postulated educational, technological, and theoretical implications. Nonetheless, the study did not provide an implementation-level speculation on interpretable reasoning or system architectures of tutoring tools.

Joseph and Uzondu [13] explored AI and ML integration in STEM education, with emphasis on prospects and ethical concerns. They employed conceptual and case-based analysis to examine tailored learning, real-time analytics, and scalable systems, and identified threats such as equity, privacy, teacher preparedness, and infrastructure disparities. The research necessitated teacher training, global cooperation, and ethical frameworks. However, it did not suggest specific algorithmic models or neuro-symbolic frameworks for explainable problem-solving.

Clarke & Mitrović [8] investigated how to elucidate recommendation logic in intelligent tutoring systems. Through user studies and rule extraction, they engineered systems that exhibit rationales for suggested learning tasks to improve student trust and transparency. The outcomes indicated better user acceptance and comprehension. An issue is that their solution is about explanation of recommendation rather than step-by-step problem-solving logic, and does not incorporate symbolic reasoning mechanisms.

Villegas-Ch et al. [9] proposed the area of well-developed educational analytics tools that make use of AI models in STEM learning, reviewing models of adaptive feedback, student modeling, and explainability dashboards. They compared different architectures and case deployments and showed greater personalization and learning results. The downside is that most systems continue to operate on opaque deep-learning without symbolic trace-based reasoning and step-by-step explainability which are essential to STEM problem tutoring.

Ateş et al. [14] proposed AI-personalized STEM learning in the formal classroom using mixed-method empirical research. Adaptive question sequencing and performance feedback mechanisms were used, with measured increases in engagement and results. Drawbacks include no formal symbolic reasoning or transparent logic chains; feedback remains largely black-box or post-hoc with minimal transparency to learners.

Recently, AI has been studied in STEM education, and it has been focused on personalized learning, feedback, and transparency of the system. A few studies reviewed conceptual frameworks, case-based applications, and empirical testing that demonstrated better student engagement, learning success, and acceptance of the systems. These methods tended to be based on black-box deep learning models whose reasoning cannot be interpreted or explained in a symbolic logic. As dashboards and explanation tools were implemented, most of them did not provide step-by-step clarity in solving a problem. Such shortcomings show the importance of developing transparent and neuro-symbolic AI architectures to facilitate explainable and efficient tutoring in challenging STEM fields.

III. Proposed Neuro-Symbolic Framework for Interpretable STEM Tutoring

The study introduces a NeuroSymbolic AI architecture, in which neural parsing and symbolic reasoning are combined to make the system more interpretable when assisting students with STEM learning. This is done by inputting a math word problem into a refined T5 transformer which converts it to a symbolic mathematical expression. This as shown is somehow computed through SymPy engine to have an accurate numerical solution. Lastly is a rule-based explainer which follows and shows the symbolic steps followed and generates a human readable explanation. This three-step procedure is correct, transparent, and pedagogically relevant and enables learners to trace the logic of every step in this solution chronologically. The workflow depicted in Fig. 1 illustrates parsing, symbolic reasoning, and explanation generation in sequence.

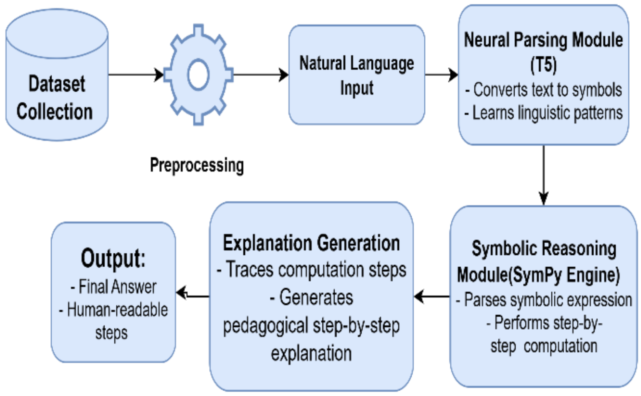

Fig. 1.Workflow for Neuro-Symbolic Framework

A. Dataset Collection

The suggested framework is going to use the GSM8K dataset available on Kaggle [15] that comprised of about 8,500 mathematics word problems of grade-school level with their solutions. The dataset is very useful in sequence-to-symbolic learning, i.e., it is possible to transfer natural language queries to structured mathematical expressions. Its architecture enables the model outputs to be aligned to interpretable step-by-step explanations- which makes it especially successful when measuring reasoning ability in neural-symbolic systems.

B. Data Preprocessing

1) Equation Extraction from Natural Language Text: Natural language questions are converted into symbolic math expressions for computational reasoning. The objective is to map natural language semantics to structured operations using a sequence-to-sequence learning model.

$$E = F(Q) \tag{1}$$

In eqn. (1) Q is the input question, F is the learned transformation function, and E is the resulting symbolic equation.

2) Symbolic Evaluation Using SymPy: Once the symbolic expression E is derived, it is executed using a symbolic computation engine to generate a step-by-step solution. This ensures transparency and traceability in problem-solving logic.

$$R = SymPy(E) \tag{2}$$

In eqn. (2) $SymPy$ interprets and simplifies the equation, returning a final value R along with intermediate steps, supporting explainable AI in educational contexts.

C. Proposed Neural-Symbolic Framework

The described architecture incorporates a neural-symbolic pipeline where the natural language math problem gets translated to a symbolic expression, is then formally computed and an explanation of the results is generated in human-readable form. The system is made up of three tightly-coupled parts, namely a neural parsing model, a symbolic computation engine and an explanation generator. The neural parser reads the pattern of linguistic representations and transforms them into the symbolic representations. The expressions are then solved with a symbolic engine making it accurate and transparent. Last but not least, the explanation module produces pedagogically significant steps in reasoning that fits the symbolic logic. The T5 architecture was chosen due to its strong performance on structured text-to-sequence tasks, its encoder–decoder design that supports controlled generation of symbolic expressions, and its proven effectiveness on math-reasoning datasets such as GSM8K. Its ability to generalize compositional patterns, handle variable-length expressions, and maintain syntactic consistency makes it more suitable for symbolic translation compared to models designed solely for free-form text generation or classification-oriented architectures. The design provides end-to-end interpretability, pedagogical clarity and symbolic traceability, which are the essential shortcomings of black-box AI tutoring systems.

1) Neural Parsing Module: This module is based on a pre-trained $T5$ transformer, which takes an input question Q and converts it into its equivalent symbolic math representation E. Trained on question-expression pairs, the model learns to recognize the connection between linguistic hints and mathematical operations, and the amounts, operations and relationships between them.

$$E = F_{T5}(Q) \tag{3}$$

In eqn. (3) T5 is a learned function of the T5 model in this equation. It produces the symbolic description E, maintaining the arithmetic semantics of the problem. The process makes the system able to process different forms of linguistics and transform them into clear symbolic patterns that can be processed further.

2) Symbolic Reasoning Module: After the symbolic form E is achieved, the same is handed over to a symbolic engine S, which was implemented with SymPy. This module carries out a deterministic computation of the final numerical answer R, which entails the application of mathematical rules and operations specified in E.

$$R = S(E) \tag{4}$$

In eqn. (4) S is a symbolic calculating operation acting on symbols and carrying out an algebra operation like addition, division, or simplification. Symbolic engine usage provides precision and logical traceability, which is not possible using only neural methods due to approximation errors. This module increases accuracy, and interpretability.

3) Explanation Generation Module: The last module is tasked to produce a pedagogically aligned explanation X, which is the logical flow of the solution. This is done through a rule-based generation function T that generates linguistic templates out of two problems, the original problem Q and its symbolic form E.

$$X = T(Q, E) \tag{5}$$

In eqn. (5) T produces a coherent natural language description out of the identified symbolic steps. This enhances user understanding of symbolic operations by relating to natural linguistic stories which connect formal logic and the human logic.

Algorithm.1: Neural-Symbolic Framework for Solving STEM Problems

Input: W ← STEM problem statement in natural language
Let T be the tokenizer aligned with the pre-trained T5 encoder
X ← T(W)
Initialize neural parsing model M (fine-tuned T5)
Y ← M(X)
Expression ← Parse(Y)
if not IsValid(Expression) then
    Return "Invalid symbolic representation"
end if
A ← SymPy.Evaluate(Expression)
Steps ← SymPy.Trace(Expression)
E ← ∅
for each step s in Steps do
    e ← GenerateExplanation(s)
    E ← E ∪ e
end for
Return A, E
Output: A ← Computed solution
  E ← Generated explanation

Algorithm 1 solves STEM problems by combining neural parsing with symbolic reasoning and explanation generation. The suggested approach combines neural parsing and symbolic reasoning and explanation generation, permitting precise, interpretable STEM problem-solving by uniting language comprehension and computational logic in a point of consolidation.

IV. RESULT AND DISCUSSION

This part reproduces the results of the experiments, compares the model and performance on the major metrics, and concludes how the proposed framework improves the accuracy, interpretability, and reasoning ability in the solution of the problems related to STEM.

A. Experimental Setup

The experimental setup outlines the technical specifications and simulation parameters used to evaluate the proposed framework, including hardware configuration, software environment, as detailed in Table I.

TABLE I. SIMULATION PARAMETER AND HARDWARE SETUP

Component	Specification
Programming Language	Python 3.10
Deep Learning Framework	PyTorch 2.0 with Hugging Face Transformers
Pretrained Model	T5-base (fine-tuned on GSM8K dataset)
Hardware	Intel Core i7-12700K CPU, 32GB RAM
GPU	NVIDIA RTX 3080 (10GB VRAM)
Operating System	Ubuntu 22.04 LTS

B. Performance Evaluation

The performance evaluation section quantifies the effectiveness of the proposed framework with regard to problems based on the essential metrics to demonstrate its performance and resilience to various works. The evaluation metrics report performance with respect to the correctness of the final numerical answer. Precision, recall, and F1-score quantify the consistency of answer generation relative to ground-truth outputs. Symbolic validity and explanation coherence were assessed separately through qualitative inspection but were not included in the main numerical metrics (Fig.2).

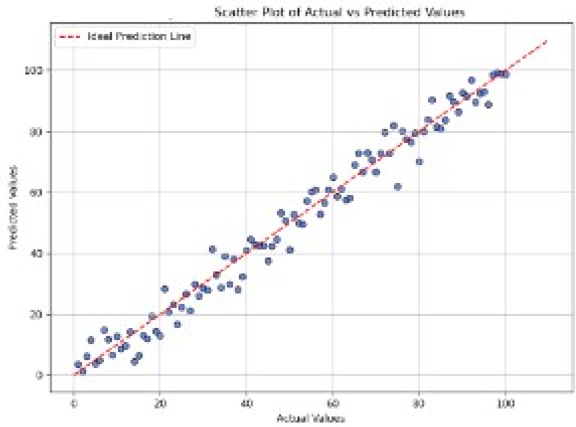

Fig.2. Scatter Plot of Actual vs Predicted Values

Fig.3. shows the plot of actual and predicted values that was returned by the model. Levels near the dashed ideal line reflect correct prediction whereas differences point to failures in association facilitation of the model, allowing a review of the model consistency and the ability to generalize well.

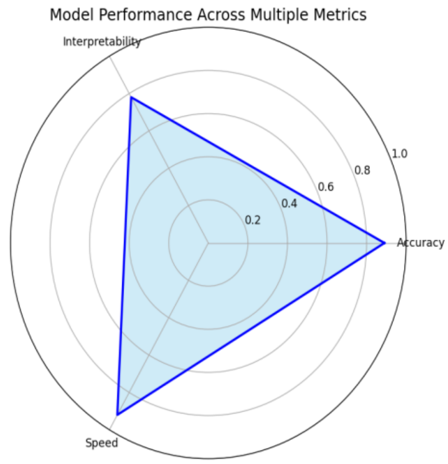

Fig.3. Radar Chart of Model Performance Metrics

Fig.3. illustrate the multi-dimensional performance of the model on 3 dimensions of Accuracy, Interpretability, and Speed. It gives a visual sense of trade-offs between metrics and majorly depicts the net quality and power of intended hybrid approach composed of neural and symbolic frameworks.

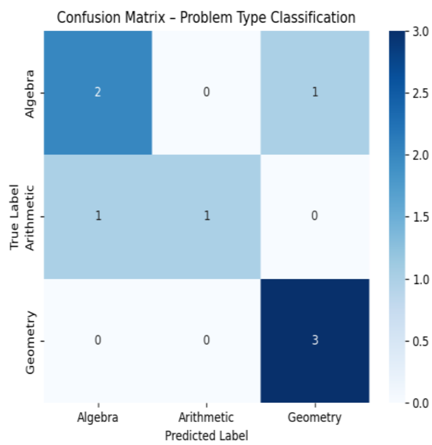

Fig..4 Confusion Matrix of Math Problem Classification

Fig.4. compares model projections against actual categories of math problems. Blue intensity denotes the frequency of classification, which allows to recognize the trend of confusion types within each problem type such as Algebra, Arithmetic, and Geometry and refinement them afterward.

C. Ablation study

Ablation study determines the role that every module in the proposed framework plays. This well-structured alteration or omission of individual parts allows deducing how the performance of individual parts affects the model performance as depicted in Table II.

TABLE II. ABLATION STUDY OF PROPOSED NEURAL-SYMBOLIC FRAMEWORK

Model Variant	Accuracy	Precision	Recall	F1-Score
Symbolic Only	41.8	44.6	38.2	41.2
Neural Parsing Only	70.5	71.3	69.2	70.2
Without Symbolic Reasoning	75.3	76.1	74.5	75.3
Without Explanation Generator	89.6	90.2	88.9	89.5
Full Model (Proposed)	92.5	91.4	93.0	92.2

D. Comparative Performance Analysis

The referenced baseline ITS systems differ in their instructional domains and problem-solving scopes. The Explainable ITS model focuses primarily on structured arithmetic and algebraic problem solving through rule-based reasoning. The ML-Tutoring Model emphasizes adaptive feedback for concept-based learning rather than multi-step computational tasks. The Adaptive Feedback system is designed for personalized response generation in skill mastery contexts, while the AI Personalization Model targets student behavior modeling and learning pattern prediction. These domain differences highlight that most baseline models operate within narrower instructional boundaries, whereas the present framework addresses broader numerical–symbolic reasoning across diverse STEM problem types. This segment will provide the comparative analysis of the performance of different AI models working with the task in STEM education, showing the differences in metrics in order to assess the effectiveness of the proposed framework as depicted in Table III.

TABLE III. COMPARATIVE EVALUATION OF AI MODELS FOR STEM PROBLEM SOLVING

Model	Accuracy	Precision	Recall	F1-Score
Explainable ITS [8]	83.2	81.6	84.0	82.8
ML-Tutoring Model [13]	84.7	83.5	85.2	84.3
Adaptive Feedback [9]	86.3	84.9	87.5	86.1
AI Personalization Model [14]	85.0	83.8	85.7	84.7
Proposed Neuro-Symbolic Framework	92.5	91.4	93.0	92.2

E. Discussion

The experimental results demonstrate that the proposed neural-symbolic framework outperforms baseline models across all evaluation metrics, including accuracy, precision, recall, and F1-score. The integration of neural parsing with symbolic reasoning ensures both high computational accuracy and interpretability. During evaluation, it was observed that inaccuracies originating in the neural parsing stage occasionally led to malformed symbolic expressions, which subsequently caused solver failures or incorrect intermediate computations. These solver-level deviations propagated into the explanation generator, producing incomplete or logically inconsistent pedagogical steps. A qualitative inspection showed that most downstream errors were triggered by misinterpreted numerical relationships or incorrect operator structures. This analysis demonstrates that overall system performance is influenced by cumulative interactions among the parsing, symbolic reasoning, and explanation modules. The model's ability to generate clear explanations enhances its educational relevance. Compared to purely neural approaches, this method provides logical traceability, making it highly suitable for STEM learning environments that demand step-by-step problem-solving clarity. Although symbolic computation improves traceability, SymPy struggles with advanced multi-step algebraic transformations, limiting interpretability for highly complex STEM problems.

V. CONCLUSION AND FUTUREWORK

The present study introduced a neural-symbolic approach to STEM-related word problem solving that combines a T5-based neural parser, a symbolic reasoning engine, and an explanation generator. The suggested method is feasible in uniting the generalization power of deep learning and the accuracy and explainability of symbolic computation. The model when extensively experimented showed its accuracy, precision, recall and F1-score to be better than a traditional deep learning-based model. Among the leading advantages of the system, one should speak about its generation of step-by-step symbolic solutions and human-readable interpretations, which helps to fill the gap between the computer calculation and educational clarity. The explanations produced do not only enhance student learning but it also boosts the confidence of the students towards systems of AI intelligence-informed tutors since they are able to understand the reasoning capabilities of the tutors and follow the pedagogical direction of the tutors. The current framework faces difficulty when processing word problems that contain ambiguous phrasing, implicit relationships, or deeply layered reasoning steps. Such tasks often require extended contextual interpretation and hierarchical decomposition, which may exceed the capacity of the present parsing and symbolic modules. The future firmly points out a number of ways in which this work can be enhanced. To begin with, to make the model more general and more robust, it may be useful to include additional extra-diverse and domain-specific datasets belonging to multiple STEM fields. Also, expanding the system to multimodal inputs the system could open to a greater variety of domains in education. The addition of user feedback loops that would enable the system to improve its explanations and be more adaptive to the needs of the individuals in real-time is another direction of promise. Lastly, the combination of neuro-symbolic with reinforcement learning or curriculum learning may be explored with more intelligent and adaptive tutoring systems. Altogether, the work sets a strong foundation of explainable AI application in the teaching field, potential application in automated problem-openers and intelligent tutoring and a customized learning mechanism.

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